Predicting d–d Transition Wavelengths of Transition-Metal Complexes:
A Machine Learning Approach with Honest Generalization Testing

Transition Metal Predictor Project | github.com/rokuromizu34/transition-metal-predictor

Abstract

Ligand field theory predicts that the wavelength of maximum d–d absorption (λ_{max}) in a transition-metal complex depends on the metal's identity, oxidation state, d-electron count, ligand field strength, and coordination geometry. This project tests whether a simple tree-based regression model, trained on tabulated λ_{max} values for 93 literature complexes, can learn this relationship well enough to generalize — not just to unseen complexes of a known metal, but to a transition metal excluded entirely from training. Standard random k-fold cross-validation reports a mean absolute error (MAE) of 68.4 nm, but a stricter leave-one-metal-out evaluation, in which each of the eight metals in the dataset is held out completely, gives an MAE of 116.0 nm — about 1.7x higher. This gap is itself the main finding: it quantifies how much of the model's apparent accuracy comes from interpolating within a familiar metal rather than truly generalizing to new chemistry, and it identifies which metals (notably Mn) are hardest to extrapolate to given the current data. A follow-up comparison shows that a simpler, regularized linear model (Ridge) generalizes better under this stricter metric (101.9 nm) than the more flexible tree ensemble, despite scoring worse under standard cross-validation — evidence that some of the tree model's apparent accuracy was overfitting rather than learned chemistry.

1. Hypothesis

H1: A regression model trained on physically motivated descriptors (metal atomic number, oxidation state, d-electron count, ligand field strength, coordination number/geometry) can predict λ_{max} for octahedral, tetrahedral, and square-planar complexes with useful accuracy.

H2 (the question this report actually investigates): Standard k-fold cross-validation overestimates the model's ability to generalize to a genuinely new transition metal, because random folds virtually always leave some complexes of every metal in the training set. A leave-one-metal-out evaluation should therefore report a meaningfully higher error than standard k-fold CV.

2. Methods

2.1 Data

93 transition-metal complexes (Co, Ni, Cu, Fe, Cr, Mn, V, Ti) with experimentally reported λ_{max} values, compiled from Miessler & Tarr, *Inorganic Chemistry* (2014). The raw table (complex formula, metal, oxidation state, ligand set, geometry, λ_{max}, color, solvent) was cleaned from an initial 107 entries to 93 by removing exact duplicates and resolving conflicting literature values.

2.2 Features

Each complex is encoded as a 10-dimensional feature vector: metal atomic number, oxidation state, d-electron count, a ligand-field-strength score (approximating the spectrochemical series), coordination number, a geometry factor, an "effective field" interaction term (ligand strength × geometry factor), and one-hot flags for octahedral / tetrahedral / square-planar geometry.

2.3 Model

ExtraTreesRegressor (scikit-learn), 200 trees, unrestricted depth, random_state=42. The same hyperparameters are used across all validation schemes below so that the comparison isolates the effect of the split strategy, not the model configuration.

2.4 Validation schemes compared

Method	Splits by	What it tests
KFold (5-fold, shuffled)	random rows	Interpolation within metals already seen in training
GroupKFold by source	literature source	Generalization across sources; metals still shared across train/test
Leave-one-metal-out (GroupKFold, k = 8)	metal	Generalization to a metal never seen in training at all

For leave-one-metal-out, GroupKFold is run with k equal to the number of unique metals (8), so each fold withholds exactly one metal's complexes entirely from training — the model predicts that metal's λ_{max} values having never seen any example of it.

3. Results

Overall MAE: standard KFold = 68.4 nm; leave-one-metal-out = 116.0 nm (≈ 1.7x higher). The gap represents the portion of apparent model accuracy that comes from interpolating within a metal the model has already seen elsewhere in training, rather than from learning a transferable relationship between electronic structure and λ_{max}.

3.1 Per-metal breakdown

Metal	Test complexes	MAE (nm)
Cr	15	33.4
V	2	44.1
Ti	3	68.7
Co	21	91.7
Ni	16	108.5
Fe	13	126.1
Cu	15	147.1
Mn	8	310.7

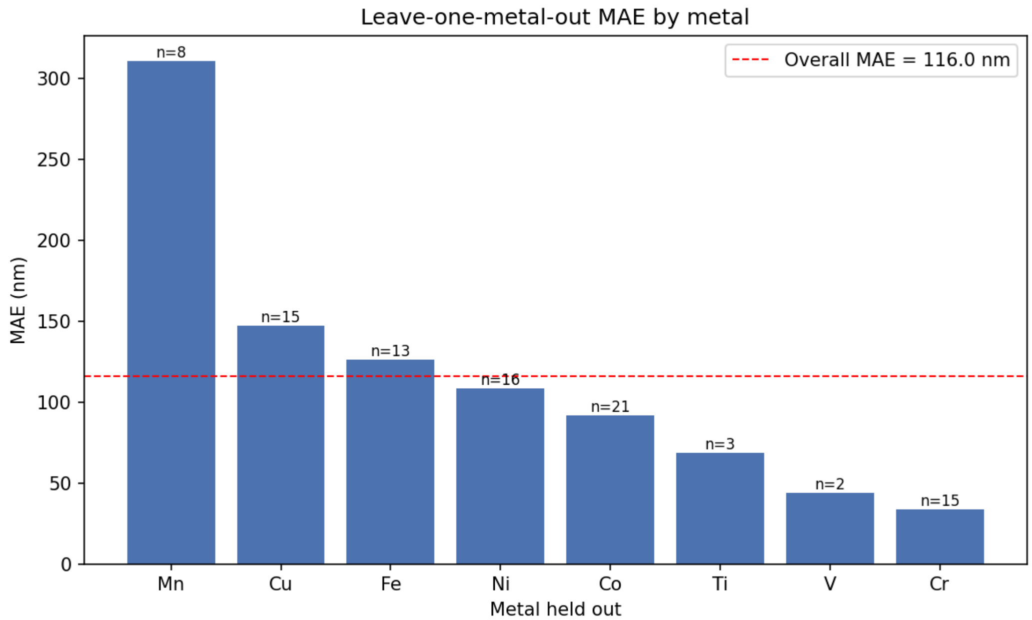

Figure 1. Leave-one-metal-out MAE by held-out metal. Dashed line: overall MAE (116.0 nm).

Cr and V are predicted comparatively well even when withheld entirely, while Mn shows by far the largest error. A plausible explanation is that Mn(II) has a d5 configuration with few close analogues among the remaining seven metals in this 93-complex dataset, so the model has little to extrapolate from once Mn examples are removed — whereas Cr(III) (d3) and neighboring metals share more overlapping electronic and geometric feature ranges.

3.2 Does model complexity affect generalization?

Tree ensembles can overfit small datasets. To test whether ExtraTreesRegressor's flexibility was actually helping or hurting generalization, the same leave-one-metal-out evaluation was repeated with two simpler linear alternatives (features standardized): plain LinearRegression and Ridge (α=1.0).

Model	KFold MAE (nm)	Leave-one-metal-out MAE (nm)
ExtraTrees (200 trees)	68.4	116.0
Ridge (α=1.0, scaled)	75.2	101.9
LinearRegression (scaled)	77.4	107.2

Ridge generalizes better than ExtraTrees under leave-one-metal-out (101.9 nm vs. 116.0 nm), despite scoring slightly worse under standard KFold (75.2 nm vs. 68.4 nm). This is consistent with overfitting: ExtraTrees' lower KFold error partly reflects memorizing metal-specific patterns rather than learning a transferable structure–property relationship, and that advantage evaporates when a whole metal is withheld. Neither model dominates for every metal individually — Ridge is markedly better for Ti (23.6 vs. 68.7 nm) while ExtraTrees is markedly better for Cr (33.4 vs. 88.4 nm) — which suggests the ideal approach may combine model classes rather than pick one universally, an avenue for future work.

4. Limitations

• **Small, single-source dataset.** All 93 complexes come from one textbook (Miessler & Tarr, 2014). This limits both statistical power and the diversity of ligand/geometry combinations the model can learn from.

• **Uneven metal representation.** V (2 complexes) and Ti (3) are barely represented; their leave-one-metal-out MAE is not statistically reliable at this sample size.

• **Hand-engineered features, not first-principles descriptors.** The ligand-field-strength score is a manual approximation of the spectrochemical series, not a computed quantum-chemical quantity (e.g. DFT-derived ligand field splitting). This caps the model's ceiling accuracy.

• **No uncertainty quantification.** ExtraTreesRegressor point predictions are reported without prediction intervals, so the model cannot currently flag when it is extrapolating outside its training distribution for a given complex.

• **Single train/test paradigm mismatch in the project roadmap.** An earlier project roadmap referenced connecting the Materials Project API to scale the dataset; that database covers DFT-computed solid-state inorganic materials (band gaps, formation energies), not molecular-complex UV-Vis spectra, so it is not a usable data source for this specific task. Dataset expansion will require further manual curation from coordination-chemistry literature.

5. Future Work

Dataset expansion. Curate additional literature tables to reach 150–200 complexes, prioritizing underrepresented metals (Mn, V, Ti) identified above as the weakest points for generalization.

Model selection under the honest metric. Section 3.2 shows Ridge generalizes better than ExtraTrees overall, but the two models disagree sharply on individual metals. A useful next step is to select or ensemble the model per prediction based on which is more reliable for chemically similar training metals, rather than deploying a single globally "best" model chosen by the flattering KFold number.

Quantum-chemical descriptors. Replace or augment the hand-built ligand-field-strength score with computed descriptors (e.g. HOMO–LUMO gap from a lightweight semi-empirical method such as xtb) for a subset of complexes, to test whether physically grounded features close the leave-one-metal-out gap.

Application to medicinal inorganic chemistry. Transition-metal complexes are directly relevant to drug design — platinum complexes (cisplatin and analogues) remain first-line chemotherapeutics, gadolinium complexes are used as MRI contrast agents, and ruthenium and copper complexes are under active investigation as next-generation anticancer agents. The same descriptor framework used here (metal identity, oxidation state, d-electron count, ligand field strength, coordination geometry) governs the electronic and, by extension, some of the reactivity and stability properties relevant to metallodrug design. A natural extension of this project is to build a small parallel dataset of therapeutically relevant metal complexes and test whether the same modeling and honest-validation approach transfers to that setting.

6. Conclusion

The model supports H1 with moderate confidence: it learns a real, physically sensible relationship between electronic/geometric descriptors and λ_{max}, achieving a 68.4 nm MAE under standard cross-validation. H2 is confirmed: leave-one-metal-out validation reveals a substantially higher, and more representative, error of 116.0 nm, showing that roughly a third of the model's apparent accuracy stems from interpolation within familiar metals rather than genuine chemical generalization. Reporting both numbers — rather than only the more flattering one — is treated here as a core part of the result, not an afterthought.

Code, data, and validation scripts: github.com/rokuromizu34/transition-metal-predictor